Paleomagnetic evidence for low-latitude glaciation during deposition of the Neoproterozoic Rapitan Group, Mackenzie Mountains, N.W.T., Canada[1]

John K. Park

Abstract: The rift-related Rapitan Group of the Mackenzie Mountains of northwestern Canada acquired several magnetizations due to pulses of hydrothermal activity. The first pulse, attributed to initiation of Rapitan rifting, produced a widespread overprint (P2) that may be reflected in the basal Mount Berg Formation. Two later pulses produced overprints similar to components found in an earlier study. Development of iron formation and hematite pigment in the overlying Sayunei Formation is attributed to the second pulse, represented by a paleopole (N = 10 sites; 334°E, 01°S; δp, δm = 4°, 9°) that coincides with poles of the Franklin igneous events of northern Canada. The Franklin episode, suggested on geological grounds to be coeval with Sayunei deposition, dates the Sayunei at ca. 725 Ma. This relation implies that rifting in Mackenzie Mountains could be related to rifting in northern Canada. A third pulse, reflected by a pole at 007°E, 16°N (N = 6 sites; δp, δm = 6°, 12°), is attributed to final rifting during deposition of the Shezal Formation at the top of the Rapitan. Overprints attributed to Sayunei and Shezal times indicate regional latitudes of 6 ± 4° and 4 ± 6° during the Sturtian glaciation. During Mount Berg time, the regional latitude could have exceeded 25°. All directions have been tilt corrected and some have been then rotated, based on comparisons with a P2 reference overprint.

Résumé : Les unités lithologiques du Groupe de Rapitan dans les monts Mackenzie du nord-ouest du Canada, dont leur dépôt est lié à un rift, ont acquis plusieurs groupes d'aimantations dues à diverses poussées d'activité hydrothermale. La première poussée hydrothermale, attribuée à la phase de l'ouverture naissante du rift associé au Groupe de Rapitan, a produit une réaimantation étendue (P2) reflétée peut-être dans l'unité de la base qui est représentée par la Formation de Mount Berg. Deux autres poussées subséquentes d'activité hydrothermale ont produit des réaimantations avec des composantes similaires à celles trouvées dans une étude précédente. Le développement de la formation ferrifère et du pigment d'hématite dans la Formation de Sayunei sus-jacente est attribuée à la deuxième poussée hydrothermale, avec un paléopôle (N = 10 sites; 334°E, 01°S; δp, δm = 4°, 9°) qui coïncide avec les pôles des intrusions de Franklin dans le nord du Canada. L'épisode intrusif de Franklin, qui selon les observations sur le terrain serait contemporain du dépôt de la Formation de Sayunei, permet d'assigner à cette dernière un âge d'environ 725 Ma. Cette corrélation implique que la formation du rift dans les monts Mackenzie peut être apparentée à la formation du rift dans le nord du Canada. Une troisième poussée d'activité hydrothermale, reflétée par un pôle à 007°E, 16°N (N = 6 sites; δp, δm = 6°, 12°), est attribuée à la phase terminale de développement du rift durant le dépôt de la Formation de Shezal qui apparaît au sommet du Groupe de Rapitan. Les réaimantations attribuées aux épisodes de Sayunei et de Shezal indiquent des latitudes régionales de 6 ± 4° et 4 ± 6° durant la glaciation sturtienne. Au moment de l'épisode de Mount Berg, la latitude régionale excédait peut-être 25°. Une correction pour inclinaison a été appliquée à chacune des directions, et pour certaines on a effectué une rotation, en se basant sur des comparaisons avec une réaimantation P2 de référence. [Traduit par la rédaction]

Introduction

The paleomagnetism of the Rapitan Group, located in the Mackenzie Mountains of northwestern Canada, was shown by Morris (1977) to be complex, with at least three definable magnetic components. Morris' study examined the problem of whether the worldwide distribution of tillites found in Neoproterozoic strata, like the Rapitan Group, indicated high- or low-latitude glaciations (e.g., Harland 1964). A recent study has proposed that these glaciations did not occur below 25° (Meert and Van der Voo 1994). A resolution of this problem is critical to the development of Neoproterozoic climate models. The present data are consistent with Morris' (1977) data, though interpreted differently, and confirm that glaciation occurred at equatorial latitudes during Rapitan time.

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Geology

Neoproterozoic rocks, extensively preserved in the Mackenzie Mountains of the northern Cordillera of Canada, occur in two major sequences: the platformal Mackenzie Mountains Supergroup (MMS) and the overlying, mainly basinal, Windermere Supergroup. Windermere strata include a passive margin phase, which ends the Proterozoic succession, and an earlier rifting phase (Fig. 1). At the base of the Windermere Supergroup, the Coates Lake and Rapitan groups comprise successive rift-related packages, separated from

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one another and from the underlying MMS by unconformities. Rifting began near the end of MMS deposition with faulting and with intrusion of widespread diabase dated at ca. 780 Ma (Jefferson and Parrish 1989; Park et al. 1995 and references therein). The widely developed unconformity at the base of the Rapitan may be due to widespread block faulting and uplift; locally, and along the basin axes (Narbonne and Aitken 1995), a conformable relation with the underlying Coates Lake Group does not suggest a large hiatus (Yeo 1981). Major deformation with thrusting and en echelon folding occurred during the Cretaceous Laramide orogeny.

Rapitan Group strata, probably deposited during the narrow ocean stage of rifting (Yeo 1981), are discontinuously exposed for about 400 km along an arcuate belt of the Mackenzie Mountains (Fig. 2). They are divided into three units: the Mount Berg, Sayunei, and Shezal formations (Fig. 1). The Mount Berg (Yeo 1978) and Shezal (Eisbacher 1978) formations include glacial and paraglacial diamictites, while the intervening Sayunei Formation contains silty rhythmites with abundant dropstones (Yeo 1981, 1984). All Rapitan formations are glaciomarine; the Sayunei is an interstadial unit, but deposited under glacial influence (Eisbacher 1981). The Mount Berg may represent a local(?) glacial advance; the Shezal Formation, a more extensive one, was laid down before a floating ice sheet (Yeo 1984). Iron formation, which probably developed synchronously in the upper Sayunei Formation and near the base of the Shezal Formation, probably has a hydrothermal origin related to volcanism and rifting (Yeo 1981).

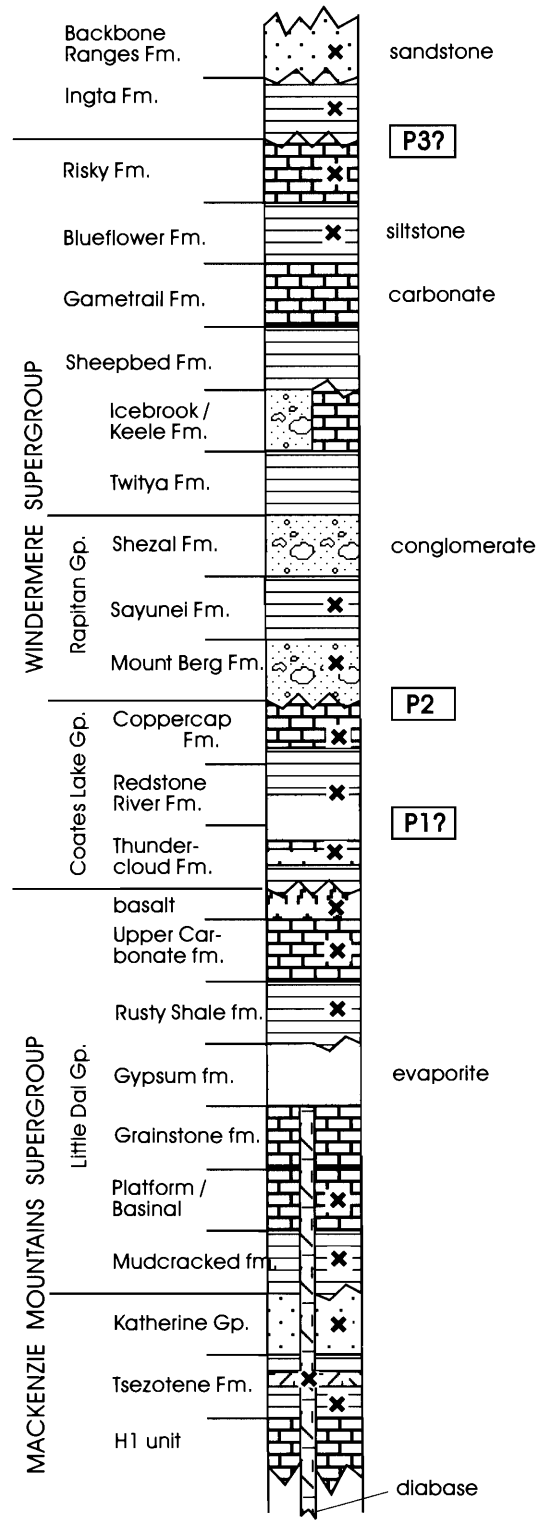
The Rapitan was sampled at three sections (Fig. 2) in the Plateau Thrust Sheet. Section 1, in the Sayunei Range near Keele River, includes sites 1–11 from the lower 200 m of the Sayunei Formation. Samples are red pebbly fine-grained sandstones and siltstones. This section is a general continuation of section 77-40 (Aitken 1981) in the Little Dal Group and of section 55 (Jefferson 1983), which continues to the top of the Coates Lake Group. Section 2 (sites 12, 13) was sampled in the upper Sayunei Formation, near the Shezal contact on the west side of Coppercap Mountain. Section 3 was sampled in the Mount Berg (site 14) and basal Sayunei formations (sites 15–17) in the Thundercloud Range to the south. It is partly equivalent to section 5 (Jefferson 1983), continuing from section 79-6 (Aitken 1981). Section 3 sites are from well-lithified fine-grained red sandstones.

Two sites were collected from the underlying Coates Lake Group: site T28 from the Redstone River Formation (section 1) and site T30 from the Thundercloud Formation (section 3). The partly bleached site T30, from the upper part of section 79-6 (Aitken 1981), overlies Thundercloud sites of Park and Jefferson (1991) and underlies sites 14–17 of this study.

Paleomagnetic methods

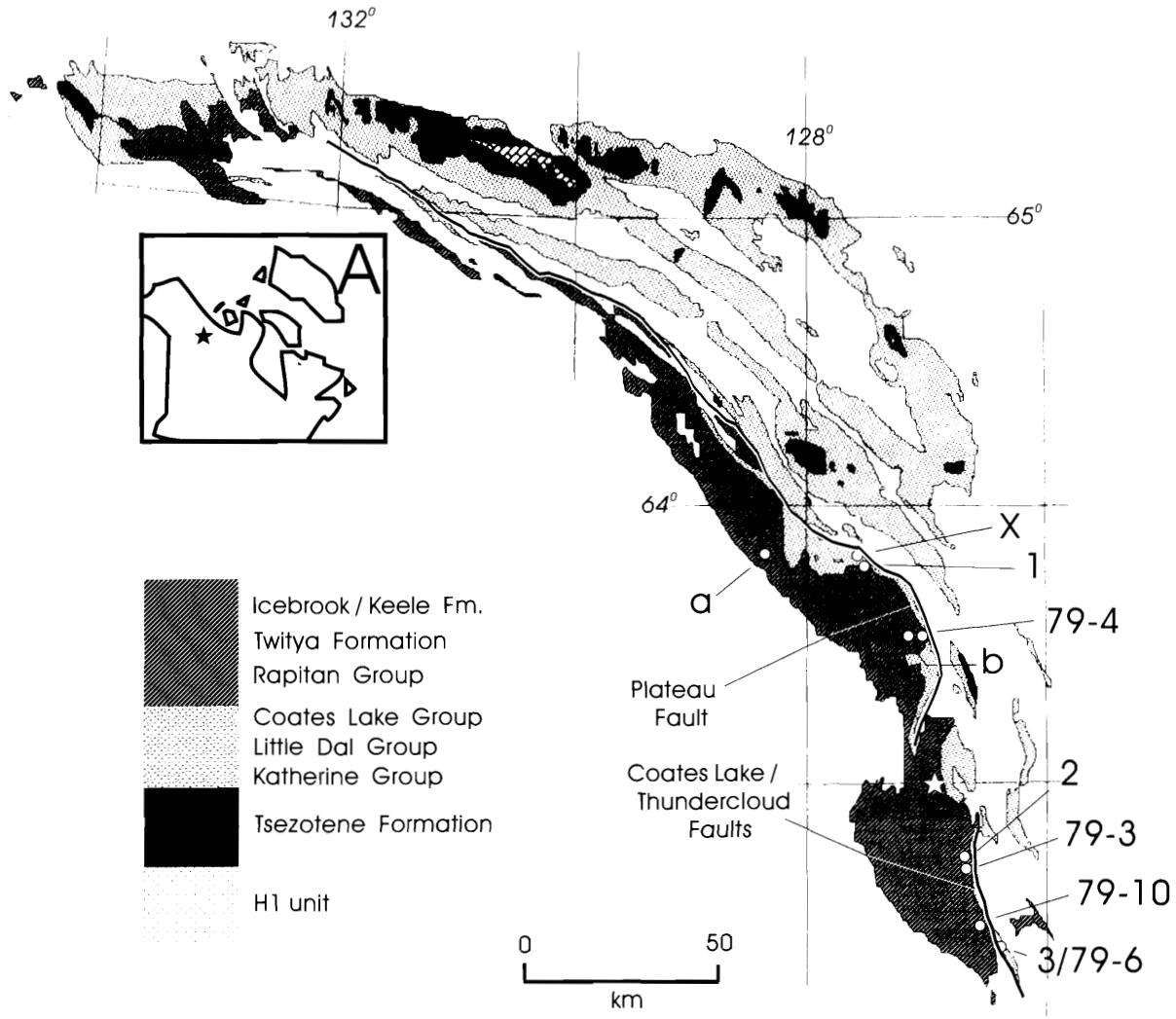
Four to six hand samples were obtained from each site; intersite stratigraphic spacings ranged from approximately 5 to over 30 m. Sample orientations were measured by sun and magnetic compasses. In the laboratory, samples were cut into several specimens (2.2 cm long, 2.5 cm in diameter). After their natural remanent magnetizations (NRM's) were measured on a Geofyzika JR-4 fast spinner magnetometer, samples were demagnetized by alternating field (AF), thermal (TH), or chemical (CH) treatments, or by several of the techniques. CH leaching was carried out using HCl acid. Details of instruments and methods may be found in Park and Jefferson (1991). Later thermal treatment of chemically leached specimens, using TSD-1 Schonstedt ovens and a

Fig. 1. Stratigraphic column of the Windermere and Mackenzie Mountains supergroups showing sampled units (crosses). Hematitization events (P1–P3) are indicated. Rock types in figure.



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Fig. 2. Generalized geological map of Neoproterozoic units in the Mackenzie Mountains of Canada showing paleomagnetic sampling localities or sections in the Rapitan Group (1–3, this study; b, Morris 1977) and Thundercloud Formation (3, this study; X, 79-4, 79-6, 79-10, Park and Jefferson 1991), and in the Little Dal basalt (a, Morris and Aitken 1982). The Coates Lake Group, shown as part of the Mackenzie Mountains Supergroup, is treated as part of the Windermere Supergroup (Narbonne and Aitken 1995) in this paper (see Fig. 1). The star indicates the reference locality (63°N, 127°W) in large- and small-scale (box A) maps for which paleolatitudes are calculated (Table 5; Fig. 12).



Geofyzika JR-4 magnetometer, was conducted in a magnetically shielded room with ambient field $<3000\ \gamma$ ($1\ \gamma \triangleq 1$ nT). Ovens maintained temperatures within $\pm 7°C$. NRM components were identified and resolved by aid from the LINEFIND (Kent et al. 1983) and Linear Spectrum Analysis (Schmidt 1982) programs, using least-squares techniques. All LINEFIND analyses have used a confidence cone of 10° (RADTOL) for defining lines, with no relaxation of linearity criteria ($\rho = 1$) and no utilization of measurement errors (α_{95} nominally set at 1°).

Results and analysis

Rapitan Group

Magnetic mineral phases

Authigenic hematite phases were difficult to distinguish under the microscope. Hematite pigment, often abundantly developed, both syn- and postdates the silica and carbonate cement. Primary precipitation of hematite and silica probably occurred throughout deposition of the Sayunei Formation (Yeo 1984). Microscopic examination of specimens leached by acid clearly showed the silica overgrowths (e.g., at sites 4, 8, 15, and 17) and unleached magnetic mineral phases. Hematite also occurs as "coarse" detrital and (or) martite grains (typically to 50 μm; up to 0.8 mm at site 14), as a pervasive micron-size intermediate phase, as euhedral grains after magnetite, and as overgrowths on quartz grains. In addition, hematite is found in detrital clasts or lithic fragments of mainly quartz and carbonate as coarse grains, overgrowths, and pigment. No magnetite was observed optically. Rutile and probable ferrian rutile partially replace hematite in about half the sites. Site 13 shows clear signs of hydrothermal alteration consistent with its stratigraphic position near the iron formation.

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Table 1. Component A from the Rapitan Group.

Si	Trt	N	n	D (°), I (°) Hor.	D (°), I (°) Paleoh.	k	α_{95} (or θ) (°)	SCo
Section 1 (127.45°W, 63.79°N)								
3	1, 5, 7	3	4	216, +30	217, +02	39	20	A2–A3
4	1, 7	2	6	222, +27	221, 00	—	(5)	A2
6	3, 6	2	3	244, +37	243, +19	—	(19)	A1
	1, 5, 7	4	10	223, +30	225, +12	91	10	A2
	3, 4	3	3	203, −01	202, −16	49	18	A3
9	6, 7	2	4	266, +17	264, +02	—	(18)	A2
	1	4	4	185, +36	191, +21	36	16	A2
10	6, 7	4	5	(253, +33)	(249, +14)	13	27	A2?
Section 2 (126.67°W, 62.71°N)								
12	1, 7	2	3	234, +67	269, −03	—	(21)	A1
13	1, 4, 7	3	3	272, +69	283, +05	25	25	A1
Section 3 (126.37°W, 62.40°N)								
14	7	2	3	294, +86	037, +28	—	(20)	A1
	1, 6, 7	4	7	264, +67	021, +42	47	14	A2
15	6	3	3	359, +83	042, +20	311	7	A1
	1	2	3	322, +75	031, +24	—	(5)	A2
	7	2	3	304, +49	002, +28	—	(16)	A2
16	1, 4, 5, 7	3	9	146, −26	350, +10[a]	222	5	A1–A3
17	1, 4, 5, 7	9	12	170, −35	007, −07[a]	11	15	A2–A3

Notes: Si, site number; Trt, treatment type: 1, CH (chemical leaching); 2, CH + AF (alternating field); 3, CH + AF + TH (thermal); 4, CH + TH; 5, AF; 6, AF + TH; 7, TH; *N*, number of samples; *n*, number of specimens. *D* and *I*, declination and inclination of the in situ (hor.) or tilt-corrected direction (paleoh.); *k*, Fisher's index of precision; α_{95} (or θ), 95% confidence error about the direction or separation between two sample directions; SCo, subcomponents. Subcomponents of A with coercive force and T_{UB} ranges over which they are resolved: A1, coarse hematite (100–660°C); A2, fine diagenetic hematite (NRM–250 mT, 100–670°C); A3, magnetite (5–50 mT, 100–580°C). Site means are calculated using sample unit weight. Where two or more subcomponents occur in a sample, they are treated as independent sample determinations of the direction. Data is rejected from further consideration where $\alpha_{95} > 25°$ or $\theta > 25°$ (directions in parentheses). Normal directions are interpreted to plot in the northeast quadrant, consistent with positioning of north poles from Neoproterozoic Laurentia in the Atlantic equatorial region.

[a]Normal direction.

Paleomagnetic components

Most specimens (87% of 257) yielded a magnetic component from at least one demagnetization technique. After components were nominated, those not meeting acceptance criteria were rejected (Table 1). Some were discarded outright, because directions are highly scattered (sites 1 and 2), skewed (site 5), or found in only one sample, or high magnetic intensities indicated that samples were probably lightning struck (site 11). In all, components from 122 specimens (47%) from 14 sites were accepted.

The next step was to classify components and then link them to magnetic phases. Magnetizations were nominally classified according to direction into shallowly inclined A (Table 1) and steeply inclined C components (Table 2; Fig. 3). These were divided into subcomponents, which were tentatively related to host magnetic mineral phases. This process involved an examination of components: their magnetic and leaching characteristics, their order of recovery during treatments, their correlation between treatments, and their tentative association with leached and unleached phases in samples. Subcomponents in the Rapitan Group and the Thundercloud Formation (below) are labeled "1" for coarse hematite, "2" for fine diagenetic hematite, or "3" for magnetite. Their unblocking temperatures (T_{UB}'s) and coercive forces are given in Tables 1 and 2. Some subcomponents are colinear (e.g., Fig. 3*c*).

CH leaching generally removes hematite phases, but usually has no affect on magnetite (Henry 1979). Fine hematite pigment is usually removed first and detrital hematite grains last (Henry 1979). Here, CH leaching apparently removed fine diagenetic hematite phases carrying A2 and C2 (Figs. 4–6), coarse hematite grains carrying A1 (Fig. 5*a*), and the micron-size hematite.

Several magnetic mineral phases were virtually unaffected by leaching. Remnant coarse hematite containing A1 (Figs. 5*a*, 6, stereograms) may be partially protected by silica cement. Resident components in this and in two phases of magnetite, which contained A3 (Figs. 5*a*, 6*b*) and C3 (Fig. 4), were

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Table 2. Component C from the Rapitan Group.

Si	Trt	N	n	D (°), I (°) Hor.	D (°), I (°) Paleoh.	k	α_{95} (or θ) (°)	SCo
Section 1 (127.45°W, 63.79°N)								
2	6, 7	3	5	(030, +48)	(029, +82)	24	26	C1?
3	5, 6, 7	2	2	049, +48	057, +76	—	(9)	C1
	1, 6, 7	3	5	346, +79	246, +66	41	19	C2
4	2, 4, 7	3	6	210, −34	205, −61	494	6	C3
	1, 7	4	9	054, +84	214, +69	47	13	C2
7	3, 4	4	4	021, +64	330, +81	144	9	C1
	1, 6, 7	5	16	013, +72	278, +80	128	7	C2
8	3, 4	3	3	(037, +67)	(016, +83)	12	38	C1?
	1, 6, 7	4	9	008, +81	253, +79	61	12	C2
10	1, 7	2	6	221, −32	218, −53	—	(12)	C3
Section 2 (126.67°W, 62.71°N)								
12	1, 6, 7	4	5	106, +67	287, +35	47	13	C2
	4, 5, 7	5	8	097, +42	302, +59	83	10	C3
13	1, 6, 7	5	8	080, +77	296, +36	35	13	C2
	4, 5, 7	5	10	084, +60	309, +51	112	7	C3

Notes: Symbols defined in Table 1. Subcomponents of C with coercive force and T_{UB} ranges over which they are resolved: C1, in coarse hematite (30−150 mT, 100−660°C); C2, in fine diagenetic hematite (100−650°C); C3, in magnetite (10−70 mT, 100−550°C).

identified by subsequent TH or AF treatment of acid-leached samples.

Some components were initially difficult to classify. This resulted because subcomponent pairs A1−C1, A2−C2, and A3−C3 have similar characteristics, which probably account for much overlap of spectra (e.g., Figs. 4−6). Also, it was not clear whether these components should be corrected for bedding tilt. Final designation of problem components is based on proximity to defined directions from arguments relating to tilt and rotation corrections. Section 2 subcomponents, C2 and A1 (Fig. 3*b*), were assigned after comparing directions before and after tilt correction with subcomponent directions of section 1 (Fig. 3*a*). At section 3, sites 16 and 17 yield subcomponents with shallow to intermediate inclinations, regardless of whether directions are tilt corrected (Fig. 3*c*). This suggests that directions should be tilt corrected, because major folding occurred only after the Earth's magnetic field became steeply directed in the Mesozoic. On the other hand, components from sites 14 and 15 of this section present a dilemma: they show shallow inclinations like A when tilt corrected, and steep inclinations like C when not. Site 15 gives three distinct subcomponents, depending on treatment type (Table 1; Fig. 3*c*). Tilt-corrected, the directions have uniform inclinations but a spread in declinations that extends to the tilt-corrected sites 16 and 17 directions (Fig. 3*c*). Thus, the tilt-corrected section 3 directions show a spread in declinations similar to that of the A directions from section 1 (Fig. 3*a*). In addition, when section 3 directions are rotated, according to evidence from the underlying Coates Lake Group at section 3 (see below), the two sets of directions coincide (see Fig. 9*a*). These directions, together with similar subcomponents from site 14, were assigned to A1 and A2.

Coates Lake Group

Paleomagnetic analysis of site T28 revealed a direction (D, I = 354°, +86°; N = 5; α_{95} = 12°) that may be related to the present Earth's field in the region or to component C found in overlying Sayunei sites (Fig. 3*a*).

Analysis of site T30 showed two magnetic components, B and D, by TH, AF, and CH treatments (Figs. 7, 8; Table 3). Subcomponents D3 and D1 reside in probable magnetite and coarse hematite, respectively, in both bleached and unbleached portions of samples. The abundant fine diagenetic hematite pigment in unbleached specimens apparently harbours D2 and B. B was also detected in bleached specimens; but due to very weak magnetic intensities (ca. 0.1 mA/m), 10−40 times less than in unbleached specimens, it could not be resolved. D2 was rarely recovered, and B never, by CH treatment, because samples crumbled after only a few hours in acid. Both, however, were revealed by TH treatment: D2 (T_{UB}'s: 200−545°C) in the previous study of section 79-6 (component B of site 23 (T23); Park and Jefferson 1991) and B (T_{UB}'s: 300−650°C) in this study (Fig. 7*b*). Present and past (Park and Jefferson 1991) D results from the Thundercloud are given in Table 3; B is given in Table 4.

Discussion

Low-inclination directions

Declinations of A components differ between sections 1 and 3 (Fig. 3). One could interpret the differences as being due to reversals of magnetization; but changing all directions to one polarity, in order to compare them, results in a very scattered and disordered array. Instead, evidence discussed below suggests that differences are largely due to local tectonic rotations. When rotation corrections are made, the array of directions assumes more order.

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Evidence for rotation

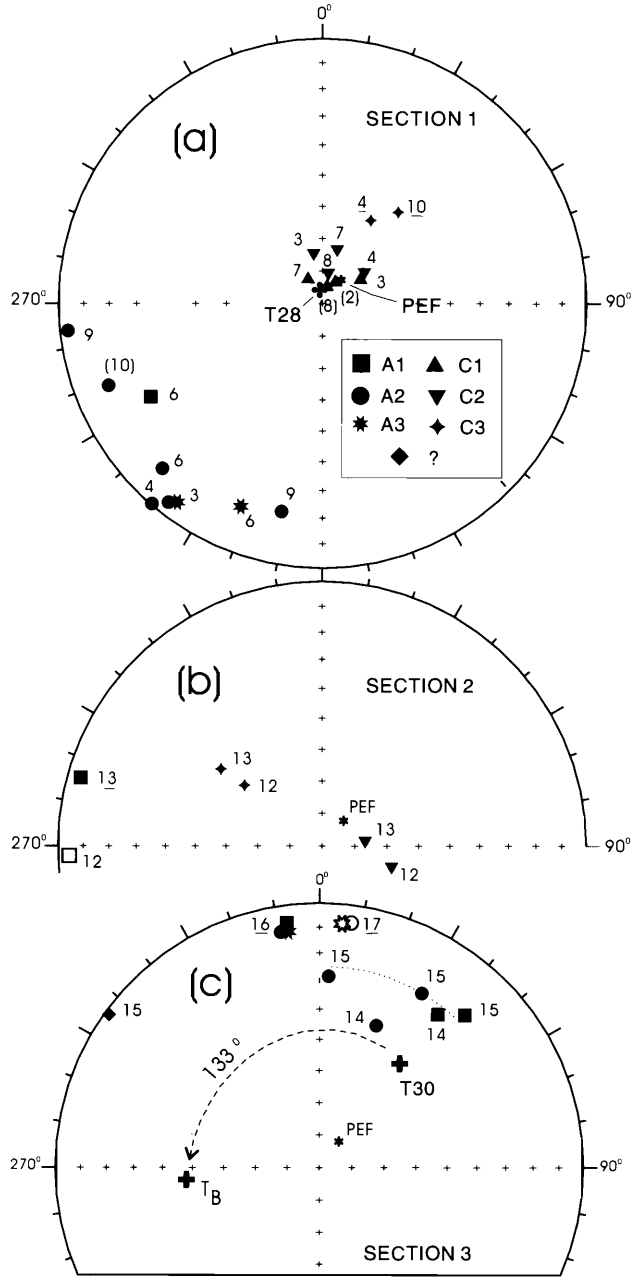
There is no direct geological evidence for terrane rotations in the study region. However, the clockwise rotations inferred from paleomagnetism (this study; Park and Jefferson 1991) are compatible with the proposed style of Neoproterozoic dextral strike-slip faulting and the "proximity of the region to a possible Proterozoic ridge-transform intersection" (Park and Jefferson, p. 24 and references therein).

Paleomagnetic evidence for terrane rotations in the Mackenzie Mountains consists of (1) clockwise rotations of at least 65° recorded in the Coates Lake Group (section 79-3, Fig. 2) (Park and Jefferson 1991), and (2) the declinational mismatch between sections 1 and 3 A directions coupled with a corresponding mismatch between an inferred B direction at section 3 and a reference B direction (Fig. 3*c*). This declinational rather than inclinational difference in directions found in this and a previous study (Park and Jefferson 1991), as well as the cohesive groupings of directions resulting from the corrective rotations (see below), suggests that tectonic rotations have occurred about largely vertical axes.

The B direction occurs as a widespread secondary overprint in fine diagenetic hematite in the Coates Lake Group (Park and Jefferson 1991) and in the underlying Mackenzie Mountains Supergroup (Morris and Aitken 1982; Park and Aitken 1986*a*, 1986*b*; Park and Jefferson 1991). The hematitization event that produced B is called P2; other similar events in the region are P1 and P3 (Park and Jefferson 1991). P2 was suggested to have occurred either at the end of Coppercap time or in the hiatus before Rapitan deposition (Park and Jefferson 1991); but in the context of this study, it may have occurred partly during Mount Berg deposition immediately following the hiatus (Fig. 1). B is found in the Plateau Thrust Sheet and in the adjacent Outer Fold Belt to the northeast, which is partly bounded by the Plateau Fault (Fig. 2) and the Mackenzie Mountains front farther northeast. Previous work suggested that some B overprints in the Plateau Thrust Sheet had undergone slight rotations (Park and Jefferson 1991). Therefore, the well-determined B direction T_B in the Tsezotene Formation of the Outer Fold Belt, which appears to be unrotated (e.g., Park and Aitken 1986*a*), is used as a reference.

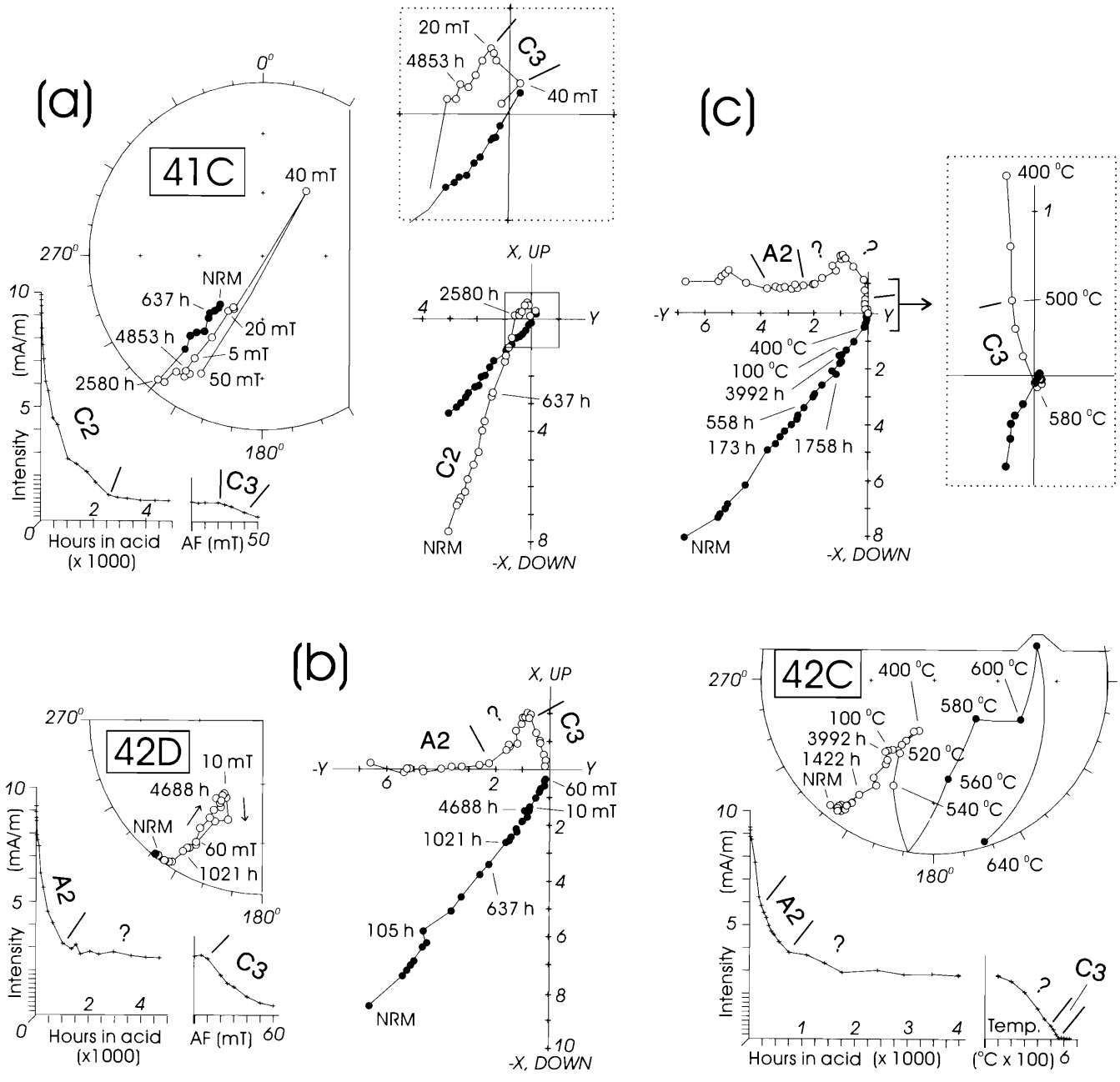
I suggest that component B of Thundercloud site T30 represents the P2 event and is suitable for comparison with the reference T_B. First, in common with other P2 components, B resides in an abundant chemically produced diagenetic phase of hematite with T_{UB}'s typically distributed between at least 300 and 625°C. Second, B has a similar inclination. Plotted *in situ* (D, I = 249°, +61°), its direction diverges (ca. 30°) from high-inclination C1–C2 directions (Fig. 3*a*) with which it shares some magnetic characteristics. Tilt-corrected, its direction is distinct from the diagenetic hematite subcomponent D2 (Fig. 8). And third, B has not been found in the Sayunei Formation. As noted, B could have formed in Mount Berg time; an A2 component with similar direction and T_{UB} range to those of B is found at site 14, above site T30 (Figs. 3*c*, 9*b*; Table 1). Because much of site T30 is bleached, B was only recovered from the unbleached specimens of two samples. However, the chemical nature and abundance of hematite pigment in the unbleached specimens suggests that B developed over a sufficient period to average out secular variation. This is

Fig. 3. Stereograms of component directions analyzed from three sections (1–3) containing Rapitan sites (a–c) (Tables 1, 2). Closed (open) symbols indicate a downward (upward) direction. PEF is the present Earth's field direction in the region. Underlined sites indicate that the direction has been reversed or has mixed polarity. Sites in parentheses indicate that the components were not used in the analysis (see text and Tables 1 and 2). Comparison of Thundercloud site T30 direction B with T_B (crosses), the reference P2 overprint direction from the Tsezotene Formation, indicates that most component directions from section 3/79-6 should be rotated counterclockwise by 133° (see text and Table 4). T28 (clover-shaped symbol) is site 28 direction from the Redstone River Formation.



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Fig. 4. Stereographic projections, intensity decay curves, and vector diagrams depicting the demagnetization of specimens from site 4 by CH and AF (a, b) or CH and TH (c) treatments. Directions are given relative to the paleohorizontal. Vectors are plotted in the horizontal $X-Y$ plane (solid symbols) and the vertical UP$-Y$ plane (open symbols). Components removed (e.g., C2) are approximately indicated. Scales are in mA/m. See Fig. 3 for further explanations.



supported by the tight grouping of directions from the eight specimens analyzed (D, I = 038°, +49°; α_{95} = 4°). Also, there is no evidence that the site has been structurally or otherwise disturbed, which is in accord with the match of D directions between site T30 and site T21, below (Fig. 10, Table 3).

Correction for rotation

Comparison of B from site T30 with T_B (Fig. 3*c*) suggests that section 3 underwent a post-Sayunei clockwise rotation of 133 ± 8°. Similar comparisons are made at all localities (Table 4). Thus, locality b (Fig. 2), carrying components X and Z of Morris (1977), appears to have rotated clockwise by 12°, based on B data from nearby section 79-4 (Park and Jefferson 1991). Section 1 directions appear not to have been rotated, as B directions in the Thundercloud Formation at section X (Park and Jefferson 1991), less than 2 km away, and in the Little Dal lavas (Morris and Aitken 1982), ~50 km to the west (locality a, Fig. 2), are within error of T_B.

After correction for rotation, Sayunei directions divide naturally into groups (Fig. 9*b*), based on the presence of

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Fig. 5. Stereographic projections, intensity decay curves, and vector diagrams depicting the demagnetization of specimens from sites 6 and 7 by CH and TH treatments. See Figs. 3 and 4 for further explanations.

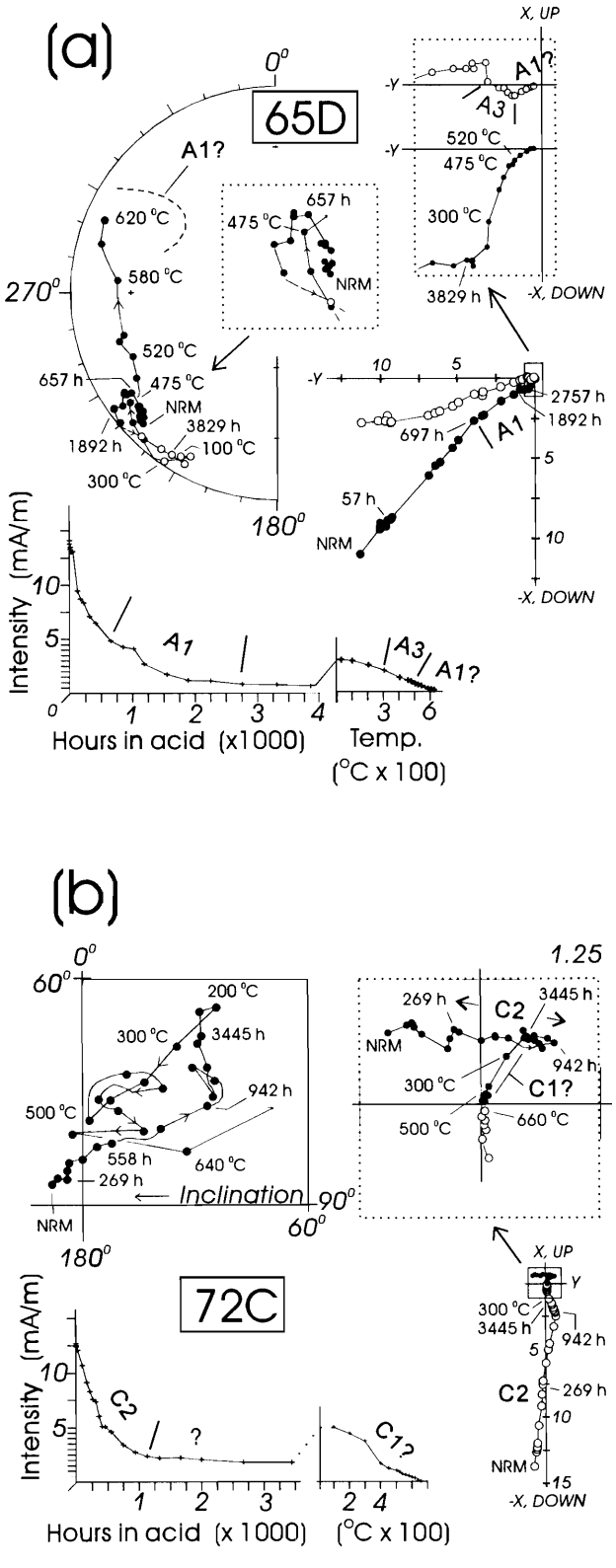
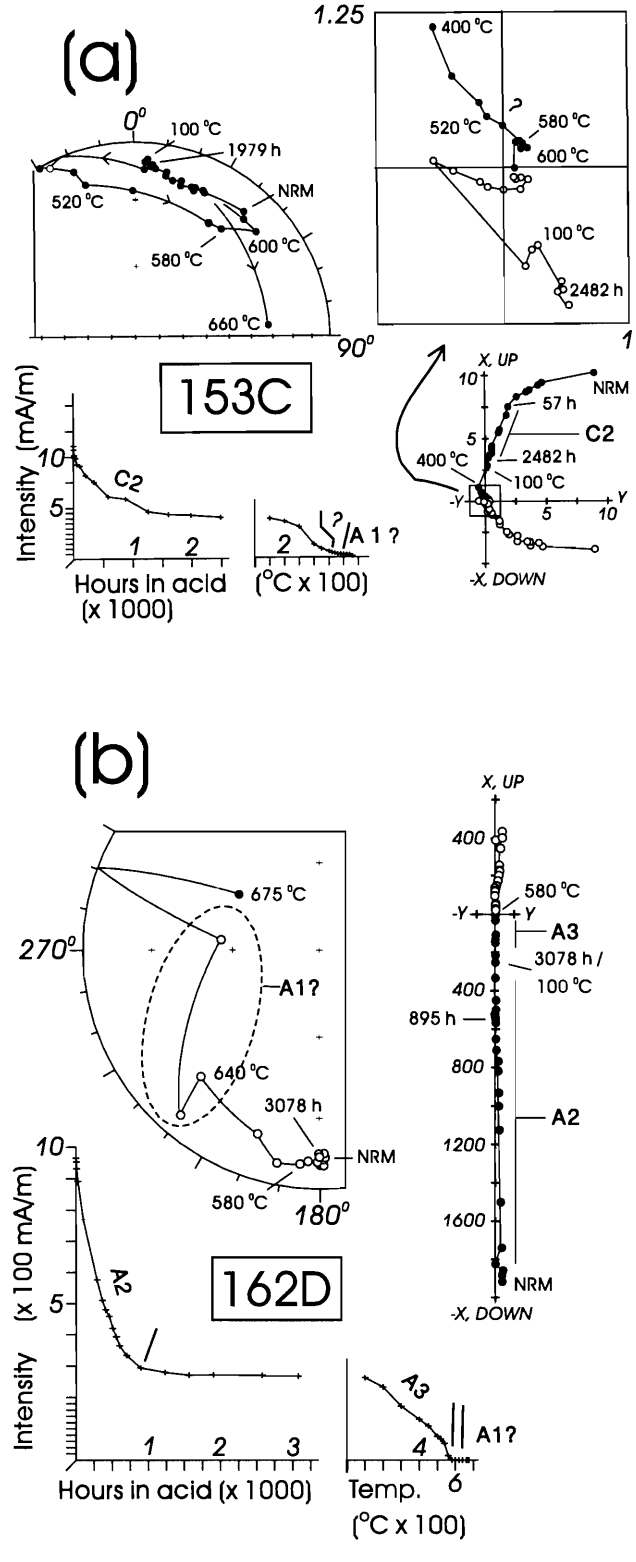


Fig. 6. Stereographic projections, intensity decay curves, and vector diagrams depicting the demagnetization of specimens from sites 15 and 16 by CH and TH treatments. Directions are not corrected for rotation. See Figs. 3 and 4 for further explanations.



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Fig. 7. Vector diagram of the AF and TH demagnetization of specimens (*a*) 301E and (*b*) 303B from Thundercloud site T30. Directions are given with respect to the *present horizontal*. Bedding data for the site: 68° dip at an azimuth of 51°. See Figs. 3 and 4 for further explanations.

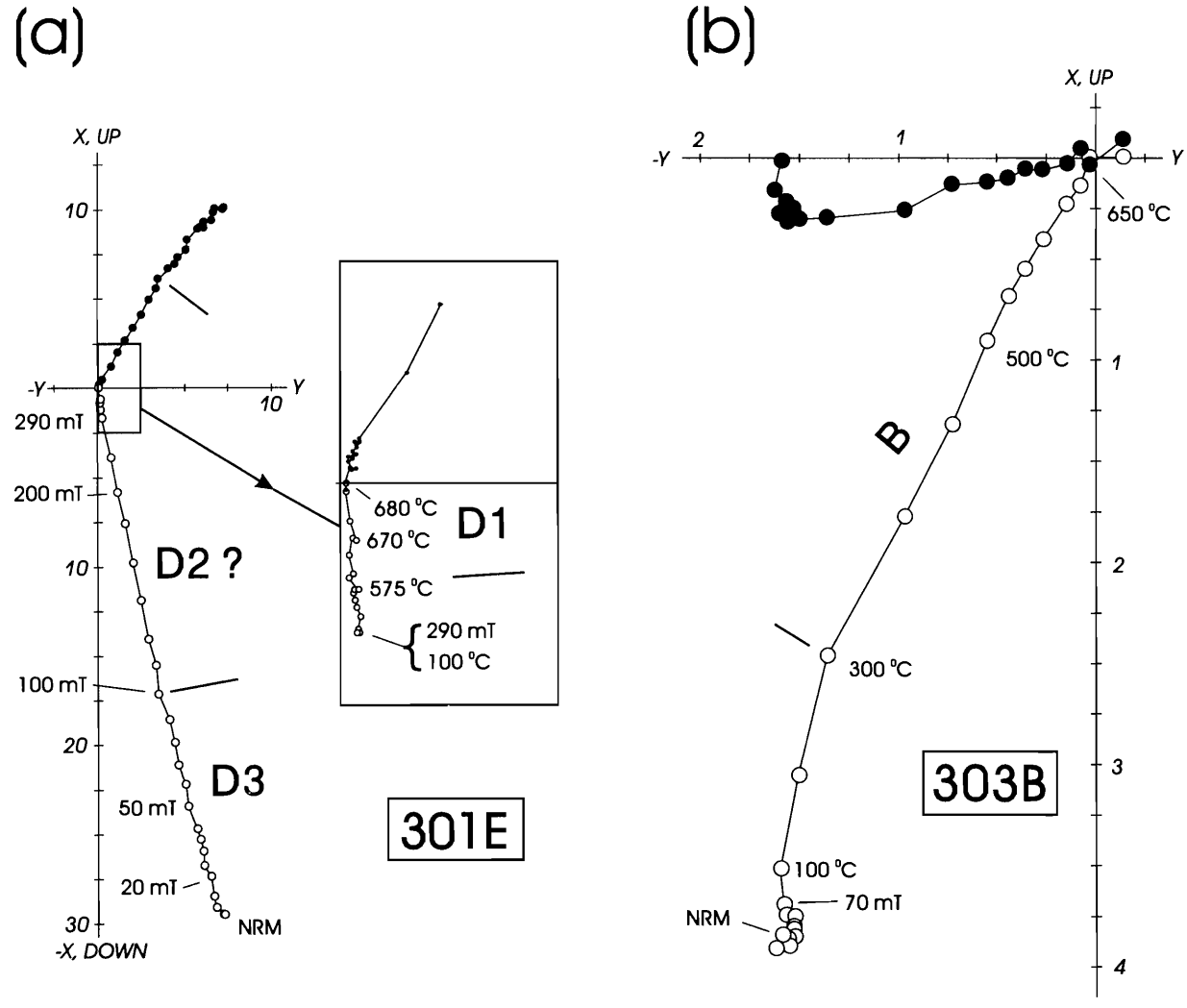
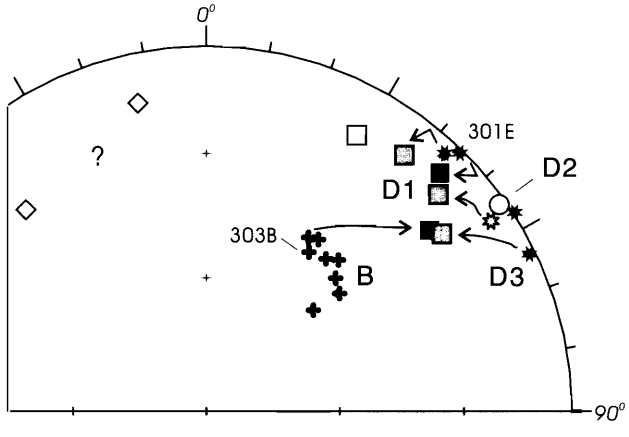


Fig. 8. Stereogram of Thundercloud site T30 component and subcomponent directions. Shaded symbols indicate that D1 was resolved from a bleached specimen. Lines connect components from the same specimen. See Fig. 3 for further explanations.



multiple subcomponents in several sites (Fig. 9*a*). Group 2 includes most A1 and several A2 subcomponents, and group 3, mostly A2 and A3 subcomponents.

Sequence of magnetization

Thundercloud Formation: Is D primary or secondary? Subcomponent D3 was earlier suggested to be primary from the presence of magnetite and associated reversals (component A in Park and Jefferson 1991). This paper shows that the D1 and D3 directions at site T23 are normal and the D3 directions at sites T21 and T30, at least 75 m above, are reverse (Fig. 10). In addition, the mixed direction of D2 in site T23 implies that the field reversed during acquisition of hematite pigment carrying D2. This scenario may support a reversal stratigraphy; but one could invoke an explanation involving secondary alteration. Further support for a primary D magnetization comes from correcting directions for the inferred 133° clockwise rotation. The resulting paleopole (TH_D)

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Table 3. Component D from the Thundercloud Formation.

Si	Section	Position °W, °N	Trt	N	n	D (°), I (°) Unrotated	D (°), I (°) Rotated	k	α_{95} (or θ) (°)	Po	SCo
T7	79-4	127.0, 63.5	7	2	3	278, +20	264, +20	—	(21)	R	D1
T8	79-4	127.0, 63.5	6, 7	2	2	(297, +09)	(283, +09)	—	(40)	R	D1
T13	79-10	126.6, 62.5	3–5, 7	6	11	262, +21	262, +21	13	20	M	D3
			1	2	3	(269, −01)	(269, −01)	—	(57)	M	D2
			1	3	4	(260, +19)	(260, +19)	20	28	M	D1
T16	X	127.5, 63.8	7	2	2	252, +05	252, +05	—	(14)	M	D3
			7	1	1	252, −06	252, −06	—	—	N	D1
T17	X	127.5, 63.8	6	1	1	252, +02	252, +02	—	—	R	D3
			4, 6, 7	1	5	260, +12	260, +12	—	—	R	D1
T21	79-6	126.4, 62.4	4	2	2	050, +11	277, +11	—	(17)	R	D3
T23	79-6	126.4, 62.4	2–4, 6	4	4	255, +05	302, −05	15	25	N	D3
			1, 7	4	5	257, +17	304, −17	11	13	M	D2
			1	4	4	258, +05	305, −05	36	15	N	D1
T30	3	126.4, 62.4	5, 7	1	5	053, 00	280, 00	—	—	R	D3
			6, 7	3	6	(041, +07)	(268, +07)	12	37	R	D1
			1	1	1	055, −02	282, −02	—	—	R	D2

Notes: Po, polarity; other symbols defined in Table 1. Original results from Park and Jefferson (1991) have been reanalyzed. Directions calculated using sample unit weight. Subcomponents of D with coercive force and T_{UB} ranges over which they are resolved: D1, coarse hematite (100–680°C); D2, fine diagenetic hematite (200–545°C); D3, magnetite (<290 mT, 100–575°C). Site means (below) are calculated using weighted subcomponent means. Acceptable site means are 264°, +20° (site T7: $N = 2$, $\theta = 21°$); 262, +21 (site T13: $N = 6$, $\alpha_{95} = 20°$); 251°, +04° (site T16: $N = 3$, $\alpha_{95} = 14°$); 256°, +07° (site T17: $N = 2$, $\theta = 13°$); 277°, +11° (site T21: $N = 2$, $\theta = 17°$); and 281°, −01° (site T30: $N = 2$, $\theta = 2°$). Outlying direction of 304°, −09° (site T23: $N = 12$, $\alpha_{95} = 11°$) is not used.

Table 4. Rotation corrections for localities.

Locality	P2 direction (D) N	D (°), I (°)	α_{95} (°)	Reference (D_0) D (°), I (°)	α_{95} (°)	Estimated rotation $D - D_0$ (°)	ΔR (°)	Rotation correction (°)
1	6	267, +48	9	263, +48	4	4	±15	0
X	1	269, +49	17[a]	263, +48	4	6	±27	0
a	5	266, +48	11	263, +48	4	3	±18	
b	5	276, +54	5	264, +48	4	12	±10	−12
79-4	5	276, +54	5	264, +48	4	12	±10	−11
2	3	276, +48	9	265, +48	4	11	±15	−11
79-3	3	276, +48	9	265, +48	4	11	±15	−11
79-10	3	265, +28	23	265, +48	4	0	±27	0
3/79-6	1	038, +49	4[b]	265, +48	4	133	±8	−133

Notes: Localities shown in Fig. 2: a, Little Dal lavas (Morris and Aitken 1982); X, 79-3, 79-4, 79-6, and 79-10, Coates Lake Group (Park and Jefferson 1991); b, Rapitan Group (Morris 1977); 1, 2, and 3 (this study). The reference direction T_B is recalculated for locality coordinates. Clockwise rotations are positive; anticlockwise, negative. Uncertainty in rotation (ΔR) from uncertainty in D (ΔD) and D_0 (ΔD_0) is $\Delta R = (\Delta D^2 + \Delta D_0^2)^{1/2}$ (Cox 1980), where $\Delta D = \sin^{-1} \sin \alpha_{95}/\cos I$ and $\Delta D_0 = \sin A_0/\sin p_0$. I is the mean paleomagnetic inclination; A_0 is the semi-angle of 95% confidence about mean pole calculated from the site poles; and p_0 is the distance between the sampling locality and the expected paleopole.

[a]Calculated from four specimens (two samples).

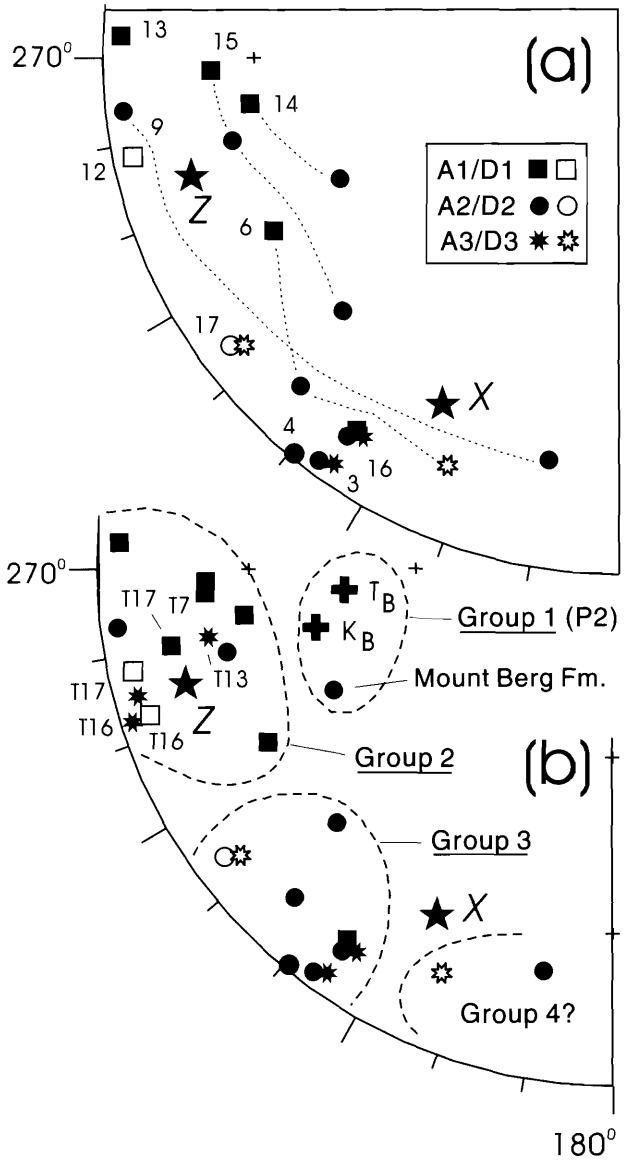
[b]Calculated from eight specimens (two samples).

from the upper sites is found to agree with the primary paleopole (S_A) from diabase intrusions that are thought to be related to basalts immediately underlying the Thundercloud Formation (Fig. 11).

Separation between directions of upper and lower sites implies that site T23 underwent a 30° clockwise rotation prior to deposition of the upper sites (Fig. 10). The two periods of clockwise rotation could correlate naturally with recorded rift tectonics in both the Coates Lake and the Rapitan periods, with the later period recording greater fault

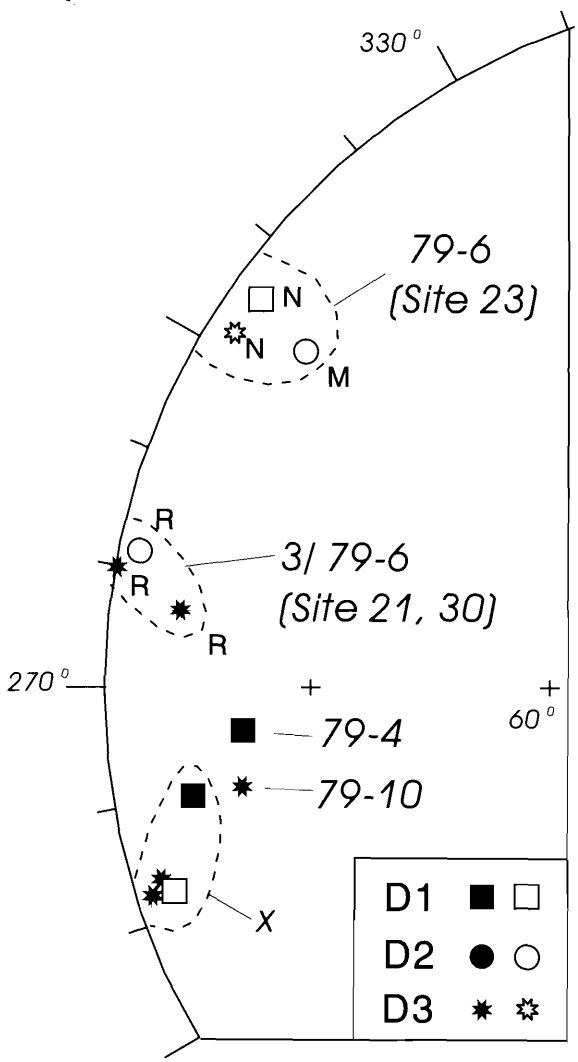
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Fig. 9. Stereographic projection of grouped directions from Sayunei and Thundercloud sites after correction for tilt and for tectonic rotations about assumed vertical axes (see text and Table 4). The X and Z components (corrected for rotation) of Morris (1977) are indicated, as well as P2 directions T_B and K_B. Rapitan directions are shown in (*a*), and combined Rapitan and Thundercloud directions in (*b*). Subcomponent directions from the same site are joined by dotted lines. See Fig. 3 for further explanations.



offsets. All D magnetizations at section 3/79-6 would therefore predate B. At the other Thundercloud sections, D, despite exhibiting reversals, is probably secondary. The close agreement of D directions at sections X, 79-4, and 79-10 with group 2 directions from the overlying Rapitan Group (Fig. 9*b*; Table 5) suggests that both sets of magnetizations were acquired concurrently, probably due to rift-related hydrothermal activity during Rapitan time. These D subcomponents, which would postdate B, are therefore included with group 2.

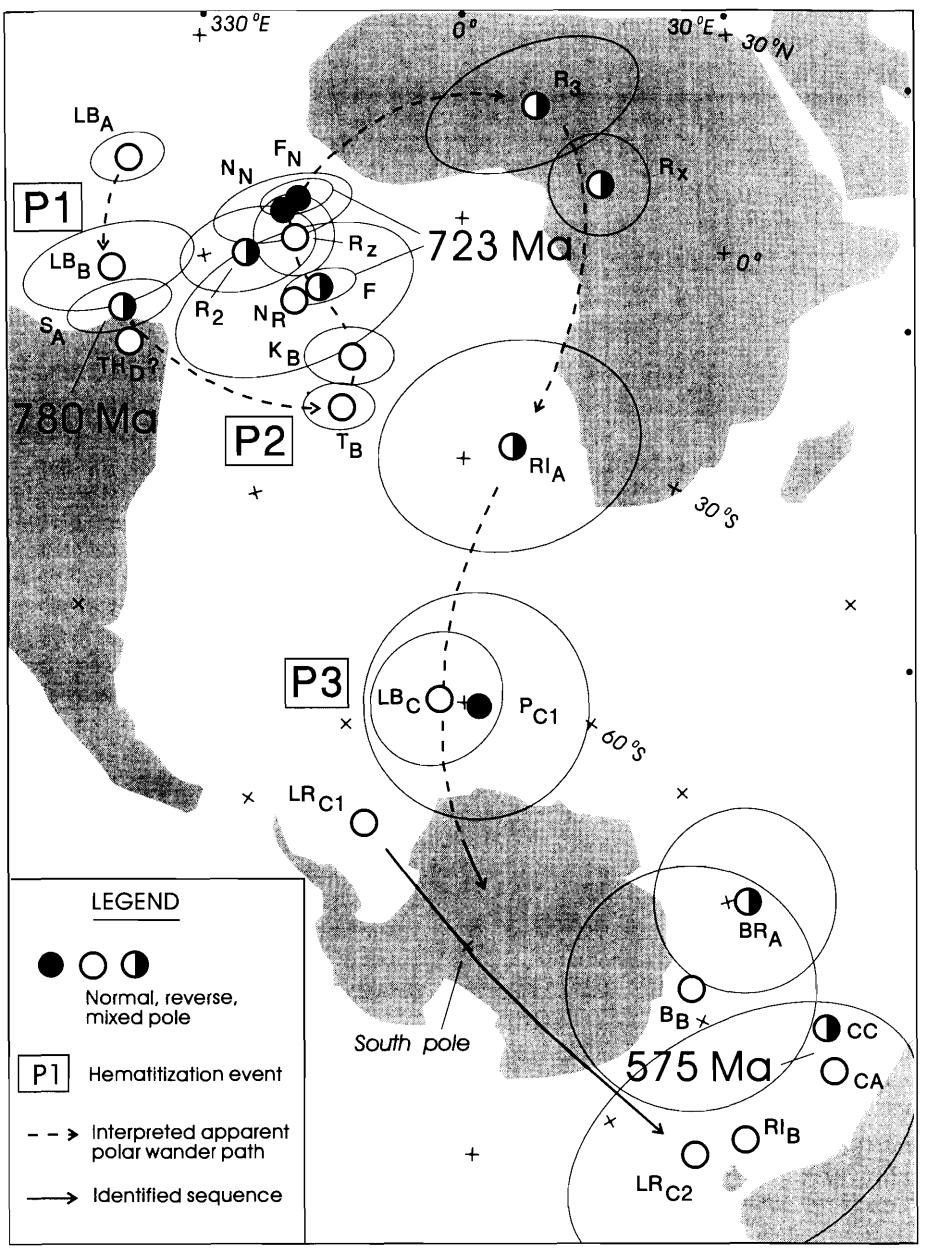
Fig. 10. Stereogram of Thundercloud D1–D3 subcomponent directions after correction for interpreted tectonic rotations about assumed vertical axes (see text and Table 4). See Fig. 3 for further explanations.



Rapitan Group: The higher inclination of the Mount Berg A2 component suggests that it could correspond to component B (group 1, Fig. 9*b*). In fact, B likely represents the initial pulse of rifting that initiated Rapitan deposition. Additional magnetizations were acquired in at least two further pulses of activity (Fig. 9). Group 2, including the secondary Thundercloud directions, represents the second pulse. It is assigned to Rapitan time from coincidence of its pole with that of the well-dated Franklin diabase, which has been estimated to be equivalent in age to the Rapitan (Rainbird et al. 1996) (Fig. 11). This coincidence could not have resulted from the effects of Franklin magmatism, for there is no evidence for Franklin diabase in the Mackenzie Mountains (Park and Rainbird 1995). Franklin diabase is widely distributed in northern Canada but in a region some 400 km to the northeast. Nor could the coincidence be due to relative rotation between the two regions (Park and Aitken 1986*a*; Park and Rainbird 1995; Park et al. 1995). The third pulse affecting the Rapitan Group produced group 3, with a more

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Fig. 11. Neoproterozoic and Early Cambrian paleopoles of the northern Cordillera. Other Laurentian paleopoles are plotted for reference. Because north poles are interpreted to plot in the Atlantic Ocean rather than in the Pacific, the usual designations of normal and reverse are interchanged. Polar errors are δp, δm (see Table 5) or $\alpha_{95}°$ (R_X, R_Z). Poles: B_B, Blueflower Formation (Park 1995); BR_A, Backbone Ranges – Ingta formations (Park 1992*b*); CA, Catoctin Formation (Meert et al. 1994); CC, Callander Complex (Symons and Chiasson 1991); F, Franklin diabase, outside the Amundsen Basin (see Park and Rainbird 1995, and references therein); F_N, Franklin diabase normal, Amundsen Basin (Park and Rainbird 1995); K_B, Katherine Group (Park and Aitken 1986*a*); LB_A, LB_B, LB_C, Little Dal Group, Basinal sequence (Park 1981*a*); LR_{C1} and LR_{C2}, Little Dal Group, Rusty Shale formation (Park and Jefferson 1991); N_N, N_R, Natkusiak basalt, normal and reverse, Amundsen Basin (see Park and Rainbird 1995, and references therein); P_{C1}, units P1 (Escape Rapids Formation) and P3 (Nelson Head Formation) of the Brock Inlier, N.W.T. (Park 1992*a*); R_2 and R_3, Rapitan Group (this study); R_X and R_Z, Rapitan Group (Morris 1977); RI_A and RI_B, Risky Formation (Park 1995); S_A, diabase of the Mackenzie Mountains (Park 1981*b*; Park et al. 1989); T_B, Tsezotene Formation (Park and Aitken 1986*b*); TH_D, Thundercloud Formation (Park and Jefferson 1991; this study).



southerly direction. Two other A subcomponents could suggest a fourth pulse. The seeming chemical nature of subcomponents of groups 2 and 3, and their distribution in both groups, implies that many of the magnetizations are secondary. Thus mean directions are probably not affected by secular variation. There is no firm evidence for the sequence of groups 2 and 3. Because group 2 is much closer to the oldest group, 1, I prefer the sequence group 1 → group 2 → group 3 (Fig. 9*b*). A 35° gap between groups 2 and 3 mean directions is unexplained; it could be largely due to apparent

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Table 5. Grand mean directions and poles from Thundercloud Formation (T) and Rapitan Group (R).

		St. cor.		Direction				Pole			
Gp	Un	Bdg	Rot	N	D (°), I (°)	k	α_{95} (or θ) (°)	Sym	°E, °S	δp (°), δm (°)	λ (°S)
Gp2	T	Yes	Yes	4	258, +13[a]	60	12	—	336, 01	6, 12	7±6
Gp2	R	Yes	Yes	6	262, +12[b]	24	14	—	333, 02	7, 14	6±7
Gp2	R, T	Yes	Yes	10	261, +12	34	8	R_2	334, 01	4, 9	6±4
Gp3	R	Yes	Yes	6	224, +07[c]	34	12	R_3	007, −16	6, 12	4±6
C1	R	Yes	No	2	025, +81	—	(16)	—	—	—	—
C2	R	No	No	6	045, +80	41	11	—	—	—	—
C2	R	No[d]	No	6	316, +83	133	6	R_{C2}	−159, −71	11, 11	−76±11
C	R	Yes	No	8	334, +84[e]	91	6	R_C	−148, −74	11, 11	−77±11

Notes: Gp, group of directions; Un, unit; St. cor., structural corrections: Bdg, bedding; Rot, rotation; Sym, symbol of pole; N, number of sites. δp, δm, semiaxes of the error of 95% confidence about the pole calculated from the mean site directions; λ, paleolatitude of the region (63°N, 127°W; see Fig. 2). See Tables 1 and 4 for further explanations.

[a]Combines sites T7, T13, T16, T17 (reverse and mixed sites, see Table 3).

[b]Combines sites 6(A1), 9(A2, first entry), 13(A1), 14(A1), 15(A1; A2, second entry).

[c]Combines sites 3, 4, 6(A3), 15(A2, third entry), 16, 17.

[d]Synfold direction, 36% unfolding.

[e]Calculated by combining C2 synfold directions with C1 directions.

polar wander (APW). It is not due to any appreciable extra clockwise rotations, as no post-Sayunei rotations are reflected in results from the underlying Thundercloud (Fig. 10). Several unclassified components may have developed during and after rotation of section 3/79-6 (Figs. 3, 8).

Fold tests (McElhinny 1964) on groups 2 and 3 directions are positive at the 99% level (group 2: N = 10 sites, k_2/k_1 = 5.34 compared with test statistic 3.13; group 3: N = 6 sites; k_2/k_1 = 15.28 compared with test statistic 4.85). A more stringent fold test (e.g., McFadden 1990) was not considered justified because of the rotational corrections employed.

High-inclination directions

C3 in magnetite has widely different directions in sections 1 and 2 (Fig. 3), and may be due to mixture with an unidentified component (Fig. 4). It is not considered further.

C2 in fine diagenetic hematite is either post- or synfolding. The synfolding direction has a minimum test statistic (ξI = 0.043, N = 6 sites) at 36% unfolding (McFadden 1990). C2 coexists with C1, attributed to coarse hematite, in at least two and perhaps four sites. A tilt test conducted on the two valid C1 directions suggests that they are prefolding, because the separation of site directions lessens (22° → 16°) on correcting for tilt and the new directions agree with C2. Close agreement between corrected C1 and C2 directions implies that C1 was acquired just before C2 during Cretaceous folding. They may be linked to a late Laramide fault that has been extrapolated through section 1 (Jefferson 1983). Its presence could explain why site 6 does not give C, whereas adjacent sites 7 and 8, perhaps closer to the fault, contain only C. C1 and C2 are averaged together in Table 3.

Comparison with Rapitan components of Morris (1977)

Morris identified three components Z, X, and Y that are respectively similar in direction to group 2, group 3 (Fig. 9*b*), and C. The Z component (D, I = 254°, +16°; N = 17 sites; α_{95} = 4°) was attributed to a thermal overprint (Morris 1977). With T_{UB}'s <500°C and low resistance to CH leaching, Z shares characteristics with some A2 subcomponents of group 2, representing primary precipitation of diagenetic hematite. This precipitation was ongoing from latest Mount Berg to early Shezal time, and probably shares the same hydrothermal origin as the iron formation (Yeo 1984). Z is found in all Morris' sites, including the iron formation and the basal Shezal Formation. Because the development of iron formation is probably coeval in both Sayunei and Shezal formations (Yeo 1984), I suggest that both Z and group 2 originated during Sayunei deposition. The mixture of magnetic phases suggests both primary and secondary remanences.

As noted, group 3 and X (D, I = 208°, +26°; N = 19 sites; α_{95} = 5°) have similar directions (Fig. 9*b*). Morris attributed X to original depositional remanence, based partly on its residence in detrital hematite. In this study, A1, which also appears to reside in detrital hematite, occurs almost exclusively in group 2, along with the similar D1, which is a secondary magnetization in the Thundercloud Formation. Because group 3 appears to be younger than group 2, X, by analogy, is probably younger than Z. This implies that X has a secondary chemical origin. I suggest that X and group 3 directions had a hydrothermal origin during final rifting in Shezal time. In my region, the subcomponents may represent thermal overprinting of preexisting magnetite and coeval development of hematite pigment. In Morris' region, X may represent the almost concurrent martitization of magnetite.

Morris found abundant evidence of a steeply inclined component Y. He ascribed it to diagenetic hematite pigment "which [appeared] to be associated with deposition of the massive conglomerates (tillites) of the Middle Rapitan" (Shezal Formation) (Morris 1977, p. 87). In my study, the similar C1−C2 component is clearly Cretaceous and perhaps later in age. The mean Y direction, corrected for tilt (D, I = 294°, +82°; N = 34 sites; α_{95} = 6°), is 5° from the mean C1−C2 direction (Table 5). Based on polarity, reverse

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and normal Y directions are 4° and 14°, respectively, from C, suggesting that Y is at least partly, if not wholly, Cretaceous to Recent in age.

Paleopoles

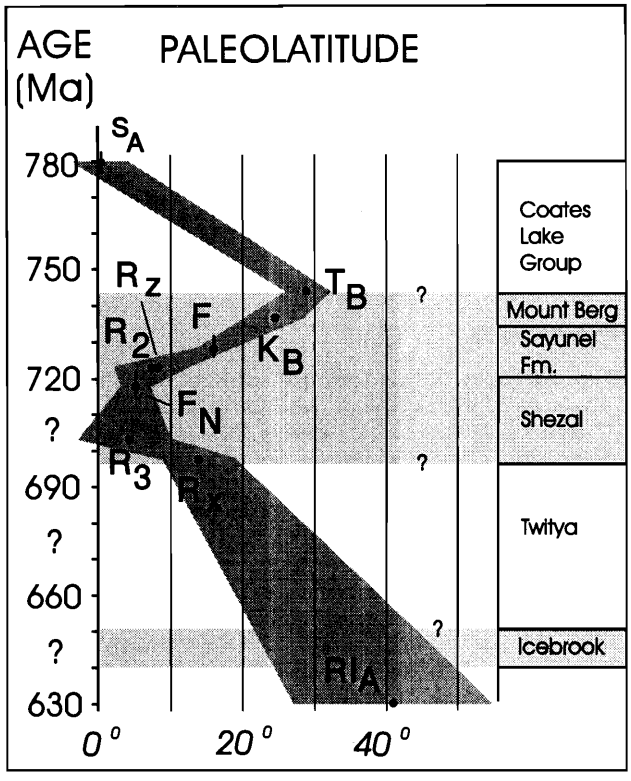
Poles T_B and K_B probably reflect the first pulse of rifting that initiated Rapitan deposition (Fig. 11). The Mount Berg A2 component could be coeval with this P2 event. Poles derived from groups 2 and 3 magnetizations, reflecting the next two pulses of rift-related hydrothermal activity during Rapitan time, are labeled R_2 and R_3, respectively (Table 5). R_2 coincides with poles from the Franklin diabase (F, F_N) and Natkusiak basalt (N_N, N_R) of the northern Arctic. Thus, the associated rifting events in these two regions may be related. The pole agreement is consistent with recent geological correlations between the Coates Lake Group of the Mackenzie Mountains and the Kilian Formation at the top of the Shaler Supergroup in the Amundsen Basin, located between about 400 and 1100 km to the northeast (Rainbird et al. 1996, and references therein). Overlying the Kilian Formation, and the intervening Kuujjua Formation in part of the basin, are the Natkusiak basalts, which are thus suggested to be coeval with the Sayunei Formation (Rainbird et al. 1996). The Natkusiak basalts and associated Franklin diabase are dated at 723^{+4}_{-2} Ma (Heaman et al. 1992). R_2 is interpreted as Sayunei in age, and coeval with the Franklin episode. A 755 ± 18 Ma U–Pb date from a granite clast from the lower Sayunei Formation provides a maximum age for the unit (G.M. Ross in Klein and Beukes 1993). The secondary pole R_3, also attributed to hydrothermal activity, probably relates to the final rifting stage during Shezal deposition.

The Neoproterozoic APW path beyond R_3 is speculative. A path over the south pole has been postulated (Park 1992*b*; 1994), ending at mid- to high-latitude Early Cambrian to latest Neoproterozoic poles (Symons and Chiasson 1991; Park 1992*b*; Meert et al. 1994). In this paper, a tentative path is drawn through RI_A, a probable primary pole from the terminal Proterozoic Risky Formation (Park 1995), and LB_C, LR_{C1}, and P_{C1}, secondary poles, possibly representing the subsequent profound unconformity (Park 1994).

Equatorial glaciation

In a recent paper, Meert and Van der Voo (1994) argued, using new paleomagnetic data and a reinterpretation of earlier data, that Neoproterozoic glaciations did not occur below 25° latitude. They questioned the reliability of data from units such as the glaciogenic Elatina Formation of Australia. However, a new study of the Elatina Formation provides strong evidence for equatorial latitudes during the Maringouin (Varanger) glaciation, dated between ca. 650 and 600 Ma (Schmidt and Williams 1995). This glaciation probably correlates with deposition of the glaciogenic Icebrook Formation (Aitken 1991) of the Windermere Supergroup. The earlier Sturtian glaciation would correspond to the Rapitan glaciation, which this study dates at ca. 725 Ma. Paleomagnetism implies that the interstadial Sayunei and the Shezal formations were under 10° latitude during development of groups 2 and 3 remanences (Fig. 12; Table 5). However, the configuration of the proposed APW path suggests that latitudes may have been higher during deposition of the Mount Berg Formation (Fig. 12); possibly greater than 25°, as implied by the probable close timing between event P2 (poles T_B and K_B) and Mount Berg deposition. Thus, contrary to Meert and Van der Voo's (1994) proposal, the evidence strongly suggests equatorial latitudes during the Sturtian glaciation.

Fig. 12. Paleolatitudes at the reference locality (63°N, 127°E; see Fig. 2) during deposition of lower Windermere units. Associated poles are shown in Fig. 11. Ages of undated poles are interpolated between 780 and 720 Ma, based on distances along the apparent polar wander path and the assumption of uniform drift of Laurentia. Ages associated with poles after pole F_N are unknown.



Conclusions

1. The Rapitan Group is dated at ca. 725 Ma from the coincidence of paleopoles from the Sayunei Formation (R_2, this study; Z, Morris (1977)) with poles of the Franklin igneous episode.
2. Glacially derived sediments of the Rapitan Group were deposited at latitudes $<10°$ during Sayunei and Shezal time, but possibly at latitudes $>25°$ during Mount Berg time.
3. Three pulses of hydrothermal activity, reflected in the paleomagnetism, are attributed to Rapitan time: the first is related to initial rifting, and the second and third, to hydrothermal activity during Sayunei and Shezal times.
4. Tectonic rotations affected Coates Lake and Rapitan units in the Neoproterozoic; they are generally corrected for by using a widespread pre-Sayunei overprint, apparently developed during initial Rapitan rifting.

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5. A major hematitization event during the Cretaceous Laramide orogeny produced a magnetization with subcomponents of pre- and synfolding age, similar to component Y identified as Neoproterozoic in age by Morris (1977).

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References

Aitken, J.D. 1981. Stratigraphy and sedimentology of the Upper Proterozoic Little Dal Group, Mackenzie Mountains, Northwest Territories. *In* Proterozoic basins of Canada. *Edited by* F.H.A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 47–71.

Aitken, J.D. 1991. Two late Proterozoic glaciations, Mackenzie Mountains, northwestern Canada. Geology, **19**: 445–448.

Cox, A. 1980. Rotation of microplates in western North America. *In* The continental crust and its mineral deposits. *Edited by* D.W. Strangway. Geological Association of Canada, Special Paper 20, pp. 305–321.

Eisbacher, G.H. 1978. Re-definition and subdivision of the Rapitan Group, Mackenzie Mountains. Geological Survey of Canada, Paper 77-35, pp. 1–21.

Eisbacher, G.H. 1981. Sedimentary tectonics and glacial record in the Windermere Supergroup, Mackenzie Mountains, northwestern Canada. Geological Survey of Canada, Paper 80-27, pp. 1–40.

Harland, W.B. 1964. Critical evidence for a great Infra-Cambrian glaciation. Geologische Rundschau, **54**: 45–61.

Heaman, L.M., LeCheminant, A.N., and Rainbird, R.H. 1992. Nature and timing of Franklin igneous events. Earth and Planetary Science Letters, **109**: 117–131.

Henry, S.G. 1979. Chemical demagnetization: methods, procedures, and applications through vector analysis. Canadian Journal of Earth Sciences, **16**: 1832–1841.

Jefferson, C.W. 1983. The Upper Proterozoic Redstone Copper Belt, Mackenzie Mountains, N.W.T. Ph.D. thesis, The University of Western Ontario, London.

Jefferson, C.W., and Parrish, R.R. 1989. Late Proterozoic stratigraphy, U–Pb zircon ages and rift tectonics, Mackenzie Mountains, northwestern Canada. Canadian Journal of Earth Sciences, **26**: 1784–1801.

Kent, J.T., Briden, J.C., and Mardia, K.V. 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence. Geophysical Journal of the Royal Astronomical Society, **75**: 593–621.

Klein, C., and Beukes, N.J. 1993. Sedimentology and geochemistry of the glaciogenic Late Proterozoic Rapitan iron-formation in Canada. Economic Geology, **88**: 542–565.

McElhinny, M.W. 1964. Statistical significance of the fold test in palaeomagnetism. Geophysical Journal of the Royal Astronomical Society, **8**: 338–340.

McFadden, P.L. 1990. A new fold test for paleomagnetic studies. Geophysical Journal International, **103**: 163–169.

Meert, J.G., and Van der Voo, R. 1994. The Neoproterozoic (1000–540 Ma) glacial intervals: No more snowball Earth? Earth and Planetary Science Letters, **123**: 1–13.

Meert, J.G., Van der Voo, R., and Payne, T.W. 1994. Paleomagnetism of the Catoctin volcanic province: A new Vendian–Cambrian apparent polar wander path for North America. Journal of Geophysical Research, **99**: 4625–4641.

Morris, W.A. 1977. Paleolatitude of glaciogenic upper Precambrian Rapitan Group and the use of tillites as chronostratigraphic marker horizons. Geology, **5**: 85–88.

Morris, W.A., and Aitken, J.D. 1982. Paleomagnetism of the Little Dal lavas, Mackenzie Mountains, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **19**: 2020–2027.

Narbonne, G.M., and Aitken, J.D. 1995. Neoproterozoic of the Mackenzie Mountains, northwestern Canada. Precambrian Research, **73**: 101–121.

Park, J.K. 1981*a*. Analysis of the multicomponent magnetization of the Little Dal Group, Mackenzie Mountains, Northwest Territories, Canada. Journal of Geophysical Research, **86**: 5134–5146.

Park, J.K. 1981*b*. Paleomagnetism of the Late Proterozoic sills in the Tsezotene Formation, Mackenzie Mountains, Northwest Territories, Canada. Canadian Journal of Earth Sciences, **18**: 1572–1580.

Park, J.K. 1992*a*. Paleomagnetism of units P1–P3 of the Late Precambrian Shaler Group, Brock Inlier, Northwest Territories. *In* Current research, part C. Geological Survey of Canada, Paper 92-1C, pp. 43–52.

Park, J.K. 1992*b*. Did Laurentia pass over the south pole during earliest Cambrian time? *In* Current research, part E. Geological Survey of Canada, Paper 92-1E, pp. 11–22.

Park, J.K. 1994. Palaeomagnetic constraints on the position of Laurentia from middle Neoproterozoic to Early Cambrian times. Precambrian Research, **69**: 95–112.

Park, J.K. 1995. Paleomagnetism of the late Neoproterozoic Blueflower and Risky formations of the northern Cordillera, Canada. Canadian Journal of Earth Sciences, **32**: 718–729.

Park, J.K., and Aitken, J.D. 1986*a*. Paleomagnetism of the Katherine Group in the Mackenzie Mountains: implications for post-Grenville (Hadrynian) apparent polar wander. Canadian Journal of Earth Sciences, **23**: 308–323.

Park, J.K., and Aitken, J.D. 1986*b*. Paleomagnetism of the Late Proterozoic Tsezotene Formation of northwestern Canada. Journal of Geophysical Research, **91**: 4955–4970.

Park, J.K., and Jefferson, C.W. 1991. Magnetic and tectonic history of the Late Proterozoic Upper Little Dal and Coates Lake Groups of northwestern Canada. Precambrian Research, **52**: 1–35.

Park, J.K., and Rainbird, R.H. 1995. Magnetic overprinting of the Neoproterozoic Shaler Supergroup, Amundsen Basin, Northwest Territories, Canada. *In* Current research, part E. Geological Survey of Canada, Paper 95-1E, pp. 113–123.

Park, J.K., Norris, D.K., and Larochelle, A. 1989. Paleomagnetism and the origin of the Mackenzie Arc of northwestern Canada. Canadian Journal of Earth Sciences, **26**: 2194–2203.

Park, J.K., Buchan, K.L., and Harlan, S.S. 1995. A proposed giant radiating dyke swarm fragmented by the separation of Laurentia and Australia based on paleomagnetism of ca. 780 Ma mafic intrusions in western North America. Earth and Planetary Science Letters, **132**: 129–139.

Rainbird, R.H., Jefferson, C.W., and Young, G.M. 1996. The early Neoproterozoic sedimentary succession B of northwestern Laurentia: correlations and paleogeographic significance. Geological Society of America Bulletin, **108**: 454–470.

Schmidt, P.W. 1982. Linearity spectrum analysis of multicomponent magnetizations and its application to some igneous rocks from south-eastern Australia. Geological Journal of the Royal Astronomical Society, **70**: 647–665.

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Schmidt, P.W., and Williams, G.E. 1995. The Neoproterozoic climatic paradox: Equatorial palaeolatitude for Maringouin glaciation near sea level in South Australia. Earth and Planetary Science Letters, **134**: 107–124.

Symons, D.T.A., and Chiasson, A.D. 1991. Paleomagnetism of the Callander Complex and the Cambrian apparent polar wander path for North America. Canadian Journal of Earth Sciences, **28**: 355–363.

Yeo, G.M. 1978. Iron-formation in the Rapitan Group, Mackenzie Mountains, Yukon and Northwest Territories. Department of Indian and Northern Affairs, Mineral Industry Report, 1975, Northwest Territories, Economic Geology Series 1978-5, pp. 170–175.

Yeo, G.M. 1981. The Late Proterozoic Rapitan glaciation in the northern Cordillera. *In* Proterozoic basins of Canada. *Edited by* F.H.A. Campbell. Geological Survey of Canada, Paper 81-10, pp. 47–71.

Yeo, G.M. 1984. The Rapitan Group: relevance to the global association of Late Proterozoic glaciation and iron-formation. Ph.D. thesis, The University of Western Ontario, London.

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